

SYDNEY GIRLS HIGH SCHOOL



2005 HSC Assessment Task 2

March 9, 2005

MATHEMATICS Extension 2

Year 12

Time allowed: 90 minutes

Topic: Complex Numbers

DIRECTIONS TO CANDIDATES:

- Attempt all questions
- Questions are not of equal value
- There are 17 questions
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used
- Write on one side of the paper only

Question 1: Given $z = 4 + 3i$,

a) find (i) iz

(ii) $\bar{z}$

(iii) $z + iz$

(iv) $z + \bar{z}$

b) plot the points represented by z , iz , $\bar{z}$, $z + iz$, $z + \bar{z}$
on an Argand Diagram

[6]

Question 2: Given $z = 1 + i$ and $w = 1 - i\sqrt{3}$, write z and w in mod-arg form and hence
or otherwise evaluate $\left(\frac{w}{z}\right)^{10}$ in the form $a + ib$

[6]

Question 3: Sketch the region on the Argand Diagram where:

$$\{3 \leq |z| \leq 5\} \cap \left\{\frac{\pi}{3} \leq \arg z \leq \frac{2\pi}{3}\right\}$$

[4]

Question 4: a) Sketch the locus of z given $\arg(z-3) = \frac{2\pi}{3}$ and

b) Find the minimum values of i) $|z|$

ii) $|z+2|$

[5]

Question 5: Solve for z in the form $a + ib$ if:

$$z^2 - z + 4 = i(z + 7)$$

[6]

Question 6: a) Given $|z-2-2i|=|z+4+6i|$,

- i) Find the equation of the locus of z
- ii) Describe this locus

b) Given $|z-2-2i|=\sqrt{2}|z+4+6i|$

- i) Find the equation of the locus of z
- ii) Describe this locus

[6]

Question 7: Assuming the result for De Moivre's Theorem for both positive and negative integers and given $z = \cos\theta + i\sin\theta$

a) Show $z^n + z^{-n} = 2\cos n\theta$

b) Express $\cos^5 \theta$ in the form $p\cos 5\theta + q\cos 3\theta + r\cos \theta$

c) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \cos^5 \theta \cdot d\theta$

[6]

Question 8: Given O is the origin and A is represented by the complex number $\sqrt{3} + i$,

a) Find C if OA is rotated through $\left(-\frac{\pi}{3}\right)$ and doubled in length to form C.

b) Find the point B which forms the parallelogram OABC

[4]

Question 9: Given w is a complex cube root of unity

a) Show that $1 + w + w^2 = 0$

b) Simplify $(1 + 2w + 2w^2)^3$

c) Find the quadratic equation which has the roots $(1 + 3w + w^2)$ and $(1 + w + 3w^2)$

[5]

Question 10: a) Given $|z + 1 - 2i| = 1$, sketch the locus of z

b) Hence or otherwise find

i) the maximum value of $\arg z$ to the nearest degree

ii) the maximum value of $|z|$

[6]

Question 11: By expanding $(\cos\theta + i\sin\theta)^4$ two different ways and by finding suitable expressions for $\cos 4\theta$ and $\sin 4\theta$, find $\tan 4\theta$ in terms of $\tan\theta$

[6]

Question 12: Find and describe the locus of z given $w = \frac{z-2}{z-2i}$ and w is a purely imaginary number.

[6]

Question 13: If A is represented by the complex number $z = 1 + 2i$ and B is represented by the complex number $w = 4 + 4i$, find the complex number D which represents the rotation of B about A by $+90^\circ$.
Hence or otherwise find the point C which completes the square $ABCD$.

[4]

Question 14: Find the four complex roots of $-8 - 8\sqrt{3}i$, giving your answers in the form $a + ib$

[6]

Question 15: Determine the locus of w , if $|z| = 1$ and $w = \frac{z-1-i}{z+1+i}$.
Describe this locus

[6]

Question 16: Find and sketch the locus of z given $\arg z^2 + \arg(-iz) = 0$

[5]

Question 17: Consider the equation $z^5 = 1$

- Solve over the complex field
- Factorise $z^5 - 1$ over the complex field
- Factorise $z^5 - 1$ over the real field
- Factorise $z^5 - 1$ over the rational field
- Show $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$
- Find the exact value of $\cos \frac{2\pi}{5}$
- Hence find the exact value of the area of the pentagon formed by the 5 complex roots of unity. [Do not attempt to simplify the answer]

[13]

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Question One (6 marks)

② $z = 4 + 3i$

(i) $iz = (4 + 3i)i = 4i + 3i^2 = -3 + 4i$

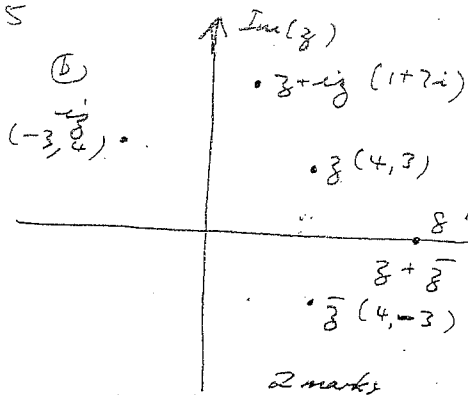
(ii) $\bar{z} = 4 - 3i$

(iii) $z + iz = 4 + 3i - 3 + 4i = 1 + 7i$

(iv) $z + \bar{z} = 4 + 3i + 4 - 3i = 8$

4 marks

Task 2
2005



Question Two

$z = 1 + i$

$z = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ 1 mark

$w = -1 - \sqrt{3}i$ 1 mark

$w = 2 \operatorname{cis} (-\frac{\pi}{3})$

4 marks

$$\begin{aligned} z^{10} &= (\sqrt{2} \operatorname{cis} \frac{\pi}{4})^{10} & w^{10} &= 2^{10} \operatorname{cis} -\frac{10\pi}{3} \\ &= 2^5 \operatorname{cis} \frac{10\pi}{4} & &= 2^{10} \operatorname{cis} \frac{2\pi}{3} \\ &= 32 \operatorname{cis} \frac{\pi}{2} & (\frac{w}{z})^{10} &= \frac{2^{10} \operatorname{cis} \frac{2\pi}{3}}{2^5 \operatorname{cis} \frac{\pi}{2}} \\ &= 32i & &= 2^5 \operatorname{cis} \frac{\pi}{6} \end{aligned}$$

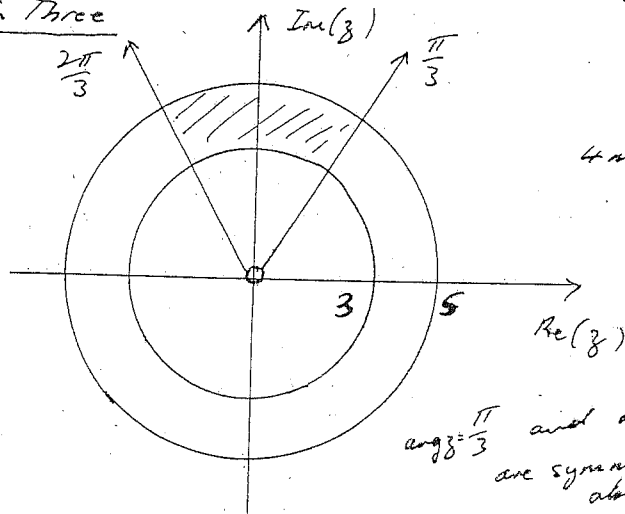
$$= 2^5 \operatorname{cis} \frac{\pi}{6}$$

$$= 32 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= 16\sqrt{3} + 16i$$

in form $a + ib$

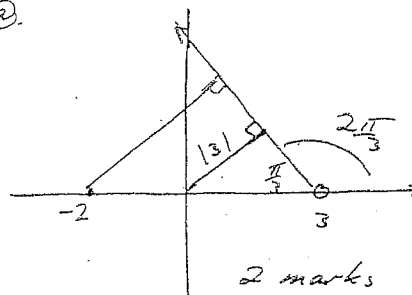
Question Three



$\arg z = \frac{\pi}{3}$ and $\arg z = \frac{2\pi}{3}$
are symmetrical
about $\operatorname{Im}(z)$ axis

Question Four (5 marks)

②



b) (i) $\min |z|$. $\frac{|3|}{3} = \sin \frac{\pi}{3}$

2 marks

$$\therefore |z| = 3 \times \frac{\sqrt{3}}{2}$$

$$|z| = \frac{3\sqrt{3}}{2}$$

(ii) $\min |z+2|$. $\frac{|3+2|}{5} = \sin \frac{\pi}{3}$

$$\therefore |z+2| = \frac{5\sqrt{3}}{2}$$

1 mark

Question Five (6 marks)

$$z^2 - z + 4 = i(z + 7)$$

$$z^2 - z + 4 = iz + 7i$$

$$z^2 - z(1+i) + (4-7i) = 0$$

$$z = \frac{(1+i) \pm \sqrt{(1+i)^2 - 4(4-7i)}}{2}$$

$$z = \frac{(1+i) \pm \sqrt{-16+30i}}{2}$$

Let $x + iy = \sqrt{-16+30i}$

$$x^2 - y^2 + 2xyi = -16 + 30i$$

$$2xy = 30 \quad xy = 15$$

$$y = 5, x = 3$$

$$\therefore x + iy = 3 + 5i$$

Now $z = \frac{(1+i) \pm (3+5i)}{2}$

$$z = 2 + 3i, z = -1 - 2i$$

Question Six

② (i) $|z - 2 - 2i| = |z + 4 + 6i|$

$$\sqrt{(x-2)^2 + (y-2)^2} = \sqrt{(x+4)^2 + (y+6)^2}$$

$$x^2 - 4x + 4 + y^2 - 4y + 4 = x^2 + 8x + 16 + y^2 + 12y + 36$$

$$\Rightarrow 0 = 12x + 16y + 44$$

$$3x + 4y + 11 = 0$$

2

(ii) locus is a straight line which is the perpendicular bisector of the interval joining (2, 2) and (-4, -6).

(b) $|z - 2 - 2i| = \sqrt{2} |z + 4 + 6i|$

$$\sqrt{(x-2)^2 + (y-2)^2} = \sqrt{2} \sqrt{(x+4)^2 + (y+6)^2}$$

$$x^2 - 4x + 4 + y^2 - 4y + 4 = 2(x^2 + 8x + 16 + y^2 + 12y + 36)$$

$$0 = x^2 + y^2 + 20x + 28y + 96$$

$$(x+10)^2 + (y+14)^2 = 200$$

$$(x+10)^2 + (y+14)^2 = 200$$

$$(x+10)^2 + (y+14)^2 = 200$$

$$(x+10)^2 + (y+14)^2 = 200$$

$$(x+10)^2 + (y+14)^2 = 200$$

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$$(x+10)^2 + (y+14)^2 = 200$$

$$(x+10)^2 + (y+14)^2 = 200$$

Question Seven / 6

a) $z = \cos \theta + i \sin \theta$

then $z^n = (\cos \theta + i \sin \theta)^n$
 $= \cos(n\theta) + i \sin(n\theta)$ ①

and $z^{-n} = (\cos \theta + i \sin \theta)^{-n}$
 $= \cos(-n\theta) + i \sin(-n\theta)$
 $\frac{1}{z^n} = \cos(n\theta) - i \sin(n\theta)$ ②

① + ② $z^n + \frac{1}{z^n} = 2 \cos(n\theta)$ 2/

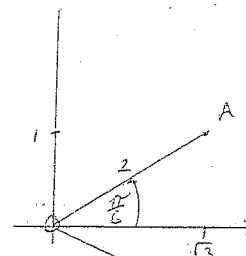
b) put $n=1$ then $z + \frac{1}{z} = 2 \cos \theta$

i.e. $(2 \cos \theta)^5 = (z + \frac{1}{z})^5$
 $= z^5 + 5z^4(\frac{1}{z}) + 10z^3(\frac{1}{z^2}) + 10z^2(\frac{1}{z^3}) + 5z(\frac{1}{z^4}) + \frac{1}{z^5}$
 $= z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$
 $= (z^5 + \frac{1}{z^5}) + 5(z^3 + \frac{1}{z^3}) + 10(z + \frac{1}{z})$
 $32 \cos^5 \theta = 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta$
 $\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{10}{16} \cos \theta$ 2/

c) $\int \cos^5 \theta d\theta = \frac{1}{80} \sin 5\theta + \frac{5}{48} \sin 3\theta + \frac{5}{8} \sin \theta$ ①

$\int_0^{\pi} \cos^5 \theta d\theta = \frac{1}{80} - \frac{5}{48} + \frac{5}{8}$
 $= \frac{8}{15}$ ① 2/

Question Eight / 4



$\sqrt{3} + i = 2 \operatorname{cis} \frac{\pi}{6}$

a) $\angle$, $2 \operatorname{cis} \frac{\pi}{6} \times 2 \operatorname{cis} (-\frac{\pi}{3})$
 $= 4 \operatorname{cis} (-\frac{\pi}{6})$
 $= 4 [\frac{\sqrt{3}}{2} - \frac{i}{2}]$
 $= 2(\sqrt{3} - i)$ 2/

b) B at $\sqrt{3} + i + 2\sqrt{3} - 2i = 3\sqrt{3} - i$ 2/

Question Nine / 5

a) $1 + w + w^2 = 0$ geometric $a=1, w=r, w^2=1$
 $S_n = \frac{a(r^n - 1)}{r - 1}$
 $= \frac{1(w^3 - 1)}{w - 1}$

$= 0$ since $w^3 = 1$ or or or 1/

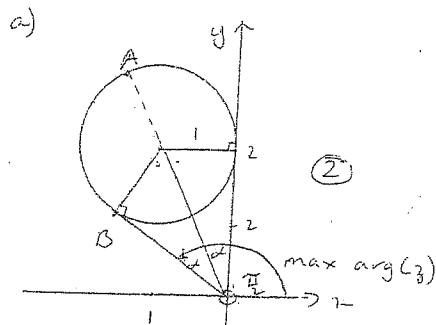
b) $(1 + 2w + 2w^2)^3 = [(1 + w + w^2) + w + w^2]^3$
 $= [(1 + w + w^2) - 1]^3$
 $= -1$ 2/

c) $\alpha = 1 + 3w + w^2$, $\beta = (1 + w + 3w^2)$
 $= [(1 + w + w^2) + 2w] = [(1 + w + w^2) + 2w^2]$
 $= 2w = 2w^2$

$\alpha + \beta = [2w + 2w^2 + 2] - 2$, $\alpha \beta = 4w^3$
 $= -2$ ① $= 4$ ①

i.e. $x^2 - (-2)x + 4 = 0$
 $x^2 + 2x + 4 = 0$ 2/

Question Ten. /6



b) 1) $\max \arg(z) = \frac{\pi}{2} + 2\alpha$
 $\alpha = \tan^{-1} \frac{1}{2}$ (at B)
 $= 26^\circ 34'$
 $\therefore \max \arg(z) = 90^\circ + 2 \times 26^\circ$
 $= 143^\circ 8'$
 (2.498 radians)

ii) $\max |z|$ at A $= \sqrt{2^2 + 1^2} + 1$
 $= \sqrt{5} + 1$

Question Eleven /6

$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$ (De Moivre's Th)
 also $(\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4\cos^3 \theta i \sin \theta + 6\cos^2 \theta i^2 \sin^2 \theta$
 $+ 4\cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta$

ii) $\cos 4\theta + i \sin 4\theta = \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6\cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$

Equate Real

$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$ ①

Equate Imaginary

$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta$ ②

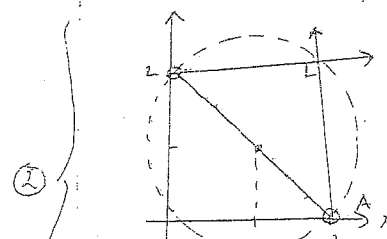
Now $\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$

$= \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta} = \frac{\cos^4 \theta}{\cos^4 \theta}$

$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$ ①

Question Twelve. /6

Geometrically



$w = \frac{z-2}{z-2i}$

$\arg w = \arg(z-2) - \arg(z-2i)$

If purely imaginary

$\arg w = \pm \frac{\pi}{2}$
 i.e. $\arg(z-2) - \arg(z-2i) = \pm \frac{\pi}{2}$ ①

Locus is a circle diameter AB, centre C(1, 1)
 Now radius $= \sqrt{1^2 + 1^2} = \sqrt{2}$ ②

i.e. $(x-1)^2 + (y-1)^2 = 2$ ②

i.e. circle centre (1, 1) radius $\sqrt{2}$ ①

excludes (0, 2) (2, 0) ①

Algebraically

$z-2i = x+iy-2i$
 $= x+i(y-2)$
 $\therefore \frac{z-2i}{z-2} = \frac{x+i(y-2)}{x-2+i(y-2)}$
 $= \frac{x}{x-2} + \frac{2i}{x-2+i(y-2)}$

$w = \frac{z-2}{z-2i} \times \frac{\bar{z}+2i}{\bar{z}+2i}$
 $= \frac{(x-2)(x+2) + 2i(x+2)}{(x-2-i(y-2))(x-2+i(y-2))}$
 $= \frac{x^2-4+2ix+4i}{(x-2)^2 + (y-2)^2}$
 $= \frac{x^2+y^2-2y-2x}{(x-2)^2 + (y-2)^2}$

Now if purely imaginary $\operatorname{Re}(z) = 0$

$x^2 + y^2 - 2y - 2x = 0$

$x^2 - 2x + y^2 - 2y = 0$

$x^2 - 2x + 1 + y^2 - 2y + 1 = 2$

$(x-1)^2 + (y-1)^2 = (\sqrt{2})^2$ ②

circle centre (1, 1) radius $\sqrt{2}$ ①

excludes (0, 2), (2, 0) ①

$$11. (\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta \text{ (de Moivre's thm)}$$

$$= \cos^4\theta + 4\cos^3\theta i\sin\theta - 6\cos^2\theta \sin^2\theta - 4\cos\theta i\sin^3\theta + \sin^4\theta$$

Equating Re & Im.

$$\sin 4\theta = 4\cos^3\theta \sin\theta - 4\cos\theta \sin^3\theta$$

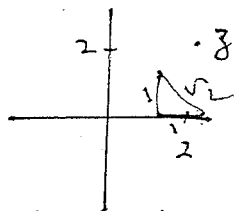
$$\cos 4\theta = \cos^4\theta - 6\cos^2\theta \sin^2\theta + \sin^4\theta$$

$$\therefore \tan 4\theta = \frac{4\cos^3\theta \sin\theta - 4\cos\theta \sin^3\theta}{\cos^4\theta - 6\cos^2\theta \sin^2\theta + \sin^4\theta}$$

$$= \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$$

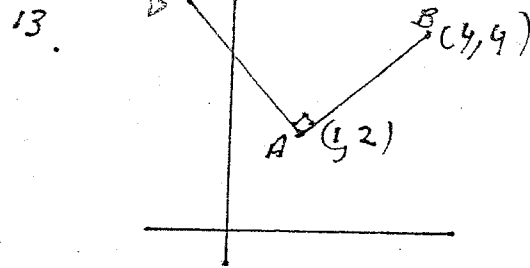
$$12. w = \frac{z-2}{z-2i}$$

if w is imaginary
 $\arg w = \frac{\pi}{2}$



z lies on a circle, diameter $(2,0)$ to $(0,2)$ from diagram

Locus is $(x-1)^2 + (y-1)^2 = \sqrt{2}$
excluding the points $(0,2)$ or $(2,0)$



$A \rightarrow B$ is $3 + 2i$
if $3 + 2i$ is rotated $\pi/2$, new point is $(3 + 2i)i = -2 + 3i$
 $\therefore A \rightarrow C$ is $A - 2 + 3i = -1 + 5i$

$$C \text{ is } B - 2 + 3i = 4 + 2i - (2 - 3i) = 2 + 5i$$

$$13. z^4 = -8 - 8\sqrt{3}i$$

$$\text{Let } z = r(\cos\theta + i\sin\theta)$$

$$\therefore z^4 = r^4(\cos 4\theta + i\sin 4\theta) = -8 - 8\sqrt{3}i \text{ (de Moivre's thm)}$$

$$|z|^4 = r^4 = 16 \quad \arg z^4 = \frac{4\pi}{3}, \frac{10\pi}{3}, \frac{16\pi}{3}, \frac{22\pi}{3}$$

$$r = 2 \quad \therefore \arg z = \frac{\pi}{3}, \frac{5\pi}{6}, \frac{4\pi}{3}, \frac{11\pi}{6}$$

$$z_1 = (\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}) = 1 + \sqrt{3}i$$

$$z_2 = (\cos \frac{5\pi}{6} + i\sin \frac{5\pi}{6}) = -\sqrt{3} + i$$

$$z_3 = (\cos \frac{4\pi}{3} + i\sin \frac{4\pi}{3}) = -1 - \sqrt{3}i$$

$$z_4 = (\cos \frac{11\pi}{6} + i\sin \frac{11\pi}{6}) = \sqrt{3} - i$$

$$15. w = \frac{z-1-i}{z+1+i}$$

$$\therefore wz + w + iw = z - 1 - i$$

$$wz - z = -w - 1 - iw - i$$

$$z(1-w) = w + 1 + i(w+1)$$

$$z = \frac{w+1+i(w+1)}{1-w}$$

$$|z|=1, \therefore \frac{|w+1+i(w+1)|}{|1-w|} = 1$$

$$\therefore |(w+1) + i(w+1)| = |1-w|$$

Let $w = x + iy$ & square both sides

$$\therefore (x+1)^2 + y^2 + (x+1)^2 + y^2 = (x-1)^2 + y^2$$

$$x^2 + 2x + 1 + y^2 + x^2 + 2x + 1 = x^2 - 2x + 1 + y^2$$

$$x^2 + 6x + 1 + y^2 = 0$$

$$(x+3)^2 + y^2 = 8$$

$\therefore$ Locus of w is a circle centre $(-3, 0)$ & radius $2\sqrt{2}$

$$\text{or } |w+3| = 2\sqrt{2}$$

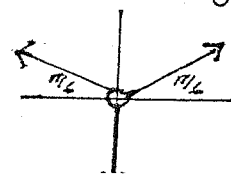
$$16. \arg z^2 + \arg(-iz) = 0$$

$$\therefore 2\arg z + \arg(-i) + \arg z = 0$$

$$\therefore 3\arg z - \pi/2 = 0$$

$$\therefore 3\arg z = \pi/2$$

$$\arg z = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$$z^5 = 1$$

$$\text{let } z = \cos \theta + i \sin \theta$$

$$\therefore z^5 = (\cos \theta + i \sin \theta)^5$$

$$= \cos 5\theta + i \sin 5\theta \quad (\text{de Moivre's})$$

$$|z| = 1, \arg z = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$z_1 = \cos 0$$

$$z_2 = \cos \frac{2\pi}{5}$$

$$z_3 = \cos \frac{4\pi}{5}$$

$$z_4 = \cos \frac{6\pi}{5}$$

$$z_5 = \cos \frac{8\pi}{5}$$

$$z^5 - 1 = (z - 1)(z - \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5})(z - \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5})(z - \cos \frac{6\pi}{5} - i \sin \frac{6\pi}{5})(z - \cos \frac{8\pi}{5} - i \sin \frac{8\pi}{5})$$

$$= (z - 1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$$

$$= (z - 1)(z^4 + z^3 + z^2 + z + 1)$$

$$z^4 + z^3 + z^2 + z + 1 = (z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$$

$$= z^4 - (2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5}) z^3 + (2 + 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5}) z^2 - \dots$$

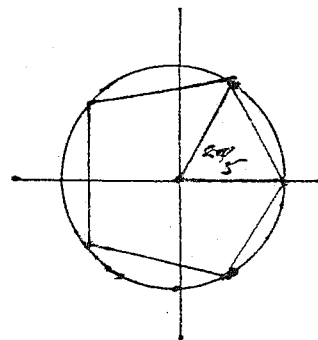
$$\text{Equate } z^3: 1 = -(2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5})$$

$$\therefore 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = -1$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$$\text{let } \frac{2\pi}{5} = \theta, \therefore \frac{4\pi}{5} = 2\theta$$

$\Rightarrow$



$$\cos 2\theta + \cos \theta = -\frac{1}{2}$$

$$\therefore 2 \cos^2 \theta - 1 + \cos \theta = -\frac{1}{2}$$

$$\therefore 2 \cos^2 \theta + \cos \theta - \frac{1}{2} = 0$$

$$4 \cos^2 \theta + 2 \cos \theta - 1 = 0$$

$$4 \cos \theta = \frac{-2 \pm \sqrt{4 + 16}}{8}$$

$$= \frac{-2 \pm 2\sqrt{5}}{8}$$

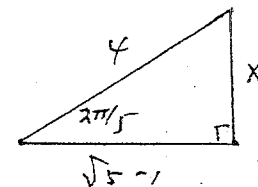
$$= \frac{-1 \pm \sqrt{5}}{4}$$

$$\text{But } \cos \frac{4\pi}{5} > 0$$

$$\therefore \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$$

$$g) A = 5 \times \frac{1}{2} ab \sin \frac{2\pi}{5}$$

$$= \frac{5}{2} \times 1 \times 1 \sin \frac{2\pi}{5}$$



$$x^2 = 16 - (\sqrt{5} - 1)^2$$

$$= 16 - (5 + 1 - 2\sqrt{5})$$

$$= 10 + 2\sqrt{5}$$

$$\therefore x = \sqrt{10 + 2\sqrt{5}}$$

$$\sin x = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

$$\therefore \text{Area is } \frac{5}{8} \sqrt{10 + 2\sqrt{5}}$$